

Brad Saund PhD

I am a Staff Applied Scientist and Robotics Engineer with 10+ years of experience building autonomous systems, robotic evaluation frameworks, and planning algorithms. I have built manipulation and autonomous driving stacks using classical and ML/RL methods, training, evaluating, and deploying in simulation and the real world.

Work Experience

2022–present **Staff Applied Scientist**, *Cruise*, San Francisco.

I built planning algorithms and tooling to help others build and evaluate planning algorithms

- I built and deployed planning algorithms to over 300 production autonomous vehicles.
- I implemented costs in C++ and CUDA using probabilistic reasoning of other agents. A sample-based planner, a trajectory optimizer consumed, and an ML Ranking network consumed these costs to output the trajectory driven by the autonomous car.
- I led a workstream pioneering a new evaluation strategy of our autonomous vehicles based on comparison to a human reference. This workstream grew the strategy from an initial concept into Cruise's primary method of end-to-end evaluation. We now run over 2 million simulations a week using this method.
- I designed and implemented the evaluation strategy that ensures our autonomous vehicle metrics are understood by our users. This has become the backbone of how Cruise/GM evaluates AV performance. I served as tech lead, initially as a sole developer and grew the team to 39 contributors, 25 of which were outside my org. This system involves C++ and Python metrics, a library for SQL queries, and a suite of frontend Streamlit dashboards.

2021–2022 **SDE II**, *AWS Robotics*, Sunnyvale.

- I founded a small team prototyping a new AWS robotics service. I conducted customer interviews, prototyped a robotic arm picking system, wrote and presented a PRFAQ to Amazon VPs.

2014–2015 **SDE**, *Amazon*, Seattle.

- I supported ad deployments to Kindle E-readers and tablets.

2012–2014 **Robotics Engineer**, *Electroimpact*, Seattle.

- I designed, built, and programmed robots that build airplanes.

Education (BS, MS, PhD in Robotics)

2017–2021 **PhD Robotics**, *University of Michigan*.

Path planning for manipulation

- Classical planning algorithms (e.g. A*, RRT and PRM variants, POMDPs, SLAM) and computer vision (e.g. Voxel Prediction CNNs)
- Planning and sensing using contact feedback
- Normalizing Flows (A predecessor of Diffusion and Flow Matching)
- LLM (GPT-3) integration with a robot planner
- Teleoperation of a dual-arm robot using a Vive and Unity

2015–2017 **MS in Robotics**, *Carnegie Mellon*.

Path planning and precision localization in confined spaces

- RL for robotic pathfinding and search
- Imitation learning

2008–2012 **BS Mechanical Engineering**, *Caltech*.

Skills

Programming ROS, TensorFlow, PyTorch, OpenCV, C++, Python

Deployment Supporting a planning stack deployed to 300 autonomous vehicles, requiring responses to SEVs

San Carlos, CA

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Selected Publications (see www.bradsaund.com for a complete list)

- 2021 **Brad Saund** and Dmitry Berenson "CLASP: Constrained Latent Shape Projection for Refining Object Shape from Robot Contact", CoRL
- 2020 **Brad Saund** and Dmitry Berenson "Diverse Plausible Shape Completions from Ambiguous Depth Images", CoRL
Brad Saund and Dmitry Berenson "Fast Planning Over Roadmaps via Selective Densification" 2020, RA-L (with ICRA presentation)
- 2019 **Brad Saund**, Sanjiban Choudhury, Siddhartha Srinivasa and Dmitry Berenson "The Blindfolded Robot: A Bayesian Approach to Planning with Contact Feedback", ISRR
- 2018 **Brad Saund** and Dmitry Berenson "Motion Planning for Manipulation with Uncertainty from Contact Sensing" 2018, ISER
- 2017 Shiyuan Chen, **Brad Saund**, and Reid Simmons "The datum particle filter: Localization for objects with coupled geometric datums" IROS
- 2017 **Brad Saund** "Planning and Localizing under Contact Uncertainty", Carnegie Mellon Master's Thesis
Alex Ansari, Julian Whitman, and **Brad Saund**. "Modular platforms for advanced inspection, locomotion, and manipulation" 2017. Waste Management Symposium
- 2016 **Brad Saund**, Shiyuan Chen, and Reid Simmons. "Touch based localization of parts for high precision manufacturing", 2016 ICRA
- 2013 **Brad Saund** and Russell DeVlieg. "High Accuracy Articulated Robots with CNC Control Systems", 2013 SAE-Aerotech

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